

Molecular Dynamics Analysis of a Polymer System

Report 1

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May 2026

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1 Introduction

Molecular dynamics simulations are widely used to study the structural and conformational behavior of polymer systems at the atomic scale. In this work, trajectory data obtained from LAMMPS simulations were analyzed to understand the flexibility, compactness, and structural organization of a polymer system.

The analyses focused on three important structural properties:

- Radius of Gyration (Rg)
- Persistence Length
- Radial Distribution Function (RDF)

The calculations were carried out using Python-based analysis scripts developed with NumPy and Matplotlib libraries. Since the trajectory files were large, computational optimizations were implemented to improve execution speed while maintaining reliable physical interpretation.

The overall objective of this work was to examine how the polymer structure evolves during simulation.

2 Objectives

The primary objectives of this work were:

1. To study the structural evolution of the polymer chain.
2. To analyze polymer compaction using Radius of Gyration.
3. To determine chain flexibility using Persistence Length.
4. To investigate local structural organization using RDF.
5. To optimize computational analysis for large trajectory datasets.

3 Simulation Data

The trajectory files analyzed were:

- Nb100Ns20gd1.0.lmptrj
- Nb100Ns20gd1.0 old.lmptrj

The simulations consisted of multiple trajectory frames containing atomic coordinates and simulation box dimensions. Frame sampling was used during analysis to reduce computational cost while preserving the important physical trends.

4 Methodology

4.1 Radius of Gyration (Rg)

The Radius of Gyration was calculated to study the overall size and compactness of the polymer chains during the simulation.

The Radius of Gyration is defined as:

$$R_g = \sqrt{\frac{1}{N} \sum_{i=1}^N (r_i - r_{cm})^2}$$

where:

- N is the total number of atoms,
- r_i is the position vector of atom i ,
- r_{cm} is the center of mass of the polymer.

The analysis was performed over multiple trajectory frames, and smoothing was applied to better visualize long-term structural trends.

Observations

The polymer initially appeared in a relatively expanded configuration. As the simulation progressed, the Radius of Gyration gradually decreased, indicating structural compaction of the polymer chains. After intermediate fluctuations and conformational rearrangements, the system approached a comparatively stable compact state.

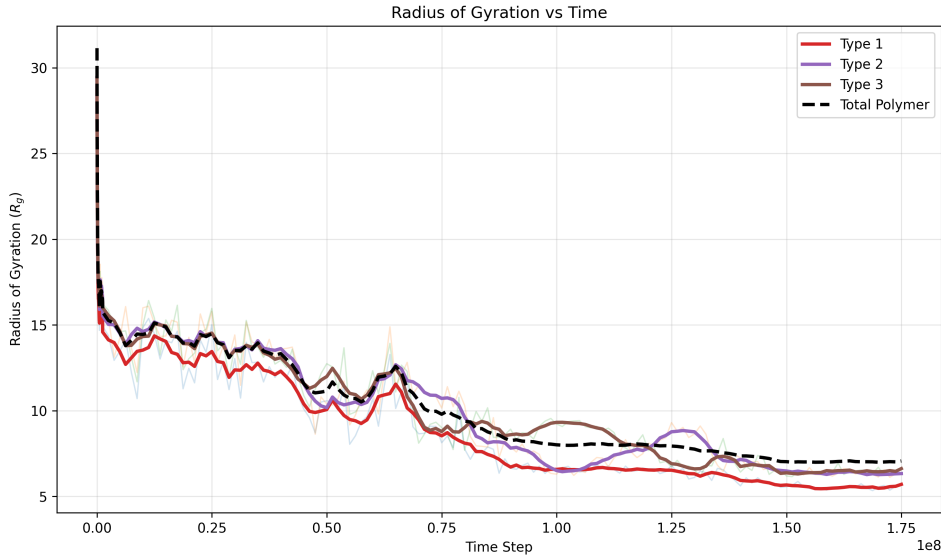


Figure 1: Radius of Gyration variation during simulation.

Interpretation

The observed decrease in R_g suggests that the polymer undergoes progressive collapse and equilibration during the simulation.

4.2 Persistence Length Analysis

Persistence length analysis was performed to quantify the flexibility of the polymer chains. The bond orientational correlation function used for the analysis was:

$$\langle t_i \cdot t_{i+s} \rangle$$

where:

- t_i represents the normalized bond vector,
- s represents the bond separation along the polymer backbone.

The persistence length was estimated by fitting the short-range orientational decay using an exponential function:

$$\langle t_i \cdot t_{i+s} \rangle \sim e^{-s/l_p}$$

where l_p is the persistence length.

Methodology

- Bond vectors were normalized before correlation calculations.
- Periodic boundary corrections were applied.
- Short-range correlations were used for fitting.
- Long-range noisy correlations were excluded.

Observations

The orientational correlation decayed rapidly with increasing bond separation and approached zero at larger separations.

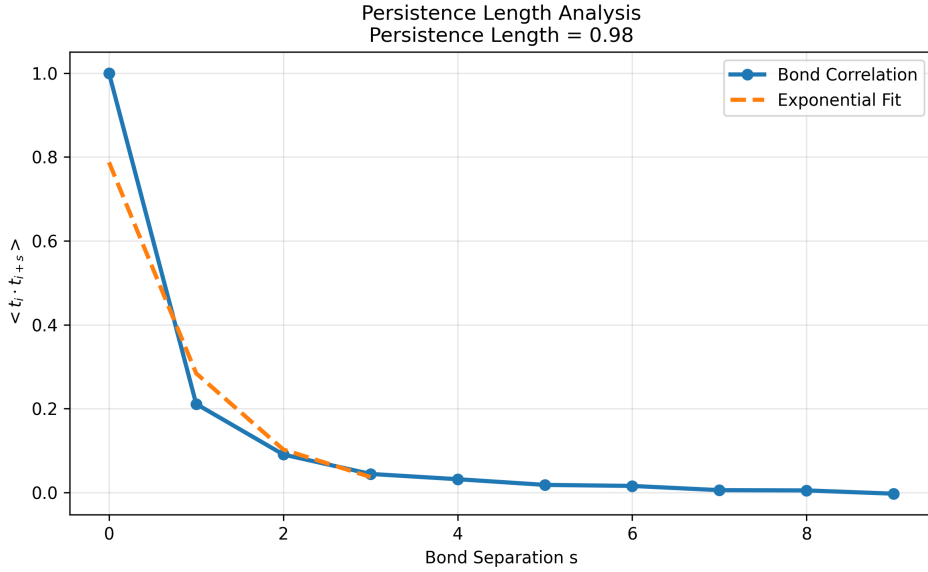


Figure 2: Bond orientational correlation function used for persistence length estimation.

Interpretation

The results indicate that the polymer chains are highly flexible and possess short orientational memory.

4.3 Radial Distribution Function (RDF)

The Radial Distribution Function (RDF) was calculated to study the local structural organization of the polymer system.

The RDF is defined as:

$$g(r) = \frac{N_{\text{pairs}}(r)}{\left(\frac{N(N-1)}{2V}\right) \Delta V(r)}$$

The analysis involved:

- pairwise distance calculations,
- periodic boundary corrections,
- shell-volume normalization,
- unique-pair normalization.

Observations

The RDF exhibited strong short-range peaks, indicating significant local structural correlations between nearby monomers. The distribution was dominated by intrachain interactions arising from bonded and neighboring monomers along the polymer chains.

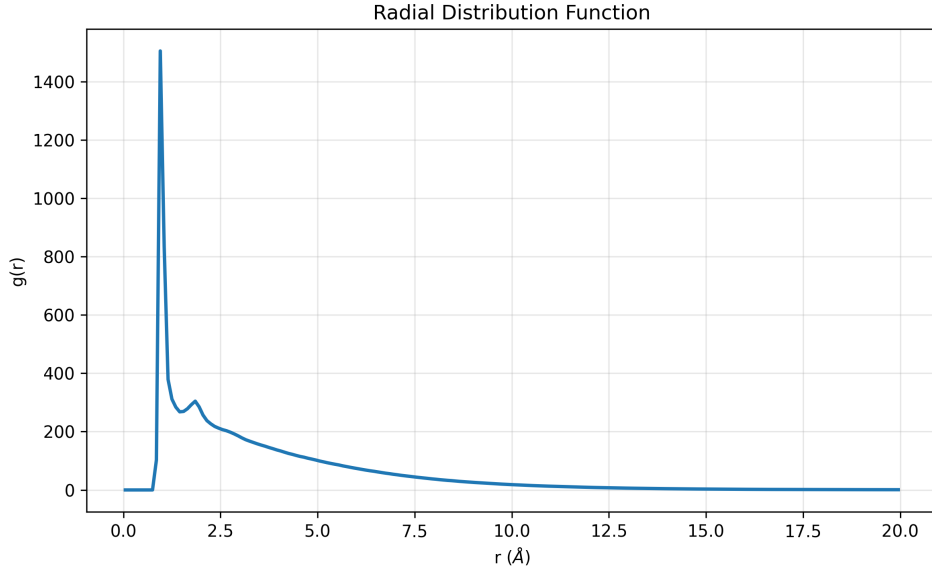


Figure 3: Radial Distribution Function of the polymer system.

Interpretation

The observed RDF behavior is consistent with compact polymer systems where chain connectivity strongly influences local structure.

5 Computational Optimization

Since the trajectory files were large, several optimizations were implemented:

- frame skipping,
- vectorized NumPy operations,
- precomputed pair indices,
- reduced memory allocations,
- use of float32 arrays where applicable.

These optimizations significantly reduced runtime while preserving the physical trends of the analyses.

6 Results Summary

Analysis	Physical Interpretation
Radius of Gyration	Polymer collapse and compaction
Persistence Length	Highly flexible polymer chains
RDF	Strong local intrachain packing

The three analyses consistently indicate:

- flexible polymer behavior,
- compact structural organization,
- gradual equilibration during simulation.

7 Conclusion

The molecular dynamics analyses successfully provided insight into the structural and conformational behavior of the polymer system.

The Radius of Gyration analysis demonstrated gradual polymer compaction over the course of the simulation. Persistence length calculations showed that the polymer chains possess high flexibility and short orientational memory. The RDF analysis revealed strong short-range structural correlations dominated by polymer connectivity and local packing effects.

The trends observed from all three analyses were physically consistent with one another and collectively describe a flexible polymer system undergoing structural compaction during simulation.

8 Software and Libraries Used

- Python
- NumPy
- Matplotlib
- LAMMPS trajectory files (.lmptj)